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Advanced Web Applications

Project Report & Documentation

Project Information

Project Name: Cloud Drive

Languages: TypeScript, HTML, CSS

GitHub Repository: <https://github.com/aahmad-1/Cloud-Drive>

Project Overview

This project develops a full-stack document storage application that lets users register, log in, and create/edit/organize their own documents in a drive. The concept is similar to that of Google Drive or any similar platform. The application supports text editing in an editor, image uploads, sharing documents with other users, downloading documents, a document recycle bin, an editing lock system that prevents simultaneous edits, and more.

Application Features & Points Requested

Basic features (mandatory requirements) ~ (25 points)

- Backend implemented with Node.js and Express
- MongoDB used for data storage
- Users can register and log in
- Authenticated users can create, edit, rename, and delete their own text/image documents
- Authenticated users can give edit permission to other authenticated users by username
- Authenticated users can share a document by a public link, giving view-only access to anyone regardless of whether they're logged in or not
- An editing lock system that checks if a document is already being edited by another user. If so, a message informs the user who is editing and prevents the user from editing
- Logout functionality
- Responsive design, usable on *most* desktop and mobile devices (see known limitations)

Additional features ~ (26 points)

Feature	Max points
Frontside framework (React)	3
WYSIWYG document editor (Quill)	2
Document can be downloaded as PDF	3
Drive shows timestamps for when a document was created and last updated	1
Documents can be sorted based on their name, date created, and last updated	1
Profile picture upload (stored on the server)	2
Toggle for dark and bright mode	1
Documents deleted are moved to a recycle bin, which can restore a document to the drive or permanently delete it from the database	2
Clone an existing document (whether it be an image or text doc)	1

Feature	Max points
User can upload personal documents to the drive (limited to image formats)	2
Translation of the whole UI into two languages (English/Finnish)	2
Search function for documents	2
Pagination to the document listing (increases documents view by a max of 7 at a time)	2
Own feature: Revoke previously given edit access from a user	1
Own feature: Direct image download (only for image docs and separate from PDF export)	1

Total requested: 25 (basic) + 26 (additional) = 51 points (one point above the 50 max in case any feature isn't credited or points are deducted)

Tech Stack

Frontend: TypeScript, Vite, React, React Router, Bootstrap, Quill, i18next, jsPDF, react-icons

Backend: TypeScript, Node.js, Express, express-validator, MongoDB, JWT, bcrypt, Multer

Installation Guide

Prerequisites:

- Node.js
- MongoDB running locally (downloading *Compass* is suggested)
- A web browser

Steps:

1. Clone the repository

- `git clone https://github.com/aahmad-1/Cloud-Drive.git` (skip if already downloaded)
- `cd Cloud-Drive`

2. Install dependencies

- `npm install`
- `cd client && npm install`
- `cd ../server && npm install`

3. Configure environment variables

- Create an `.env` file inside the *server* folder and add the following:
 - `SECRET=secret_key` (can be anything you want)
 - `PORT=3000`
 - `MONGO_URI=mongodb://127.0.0.1:27017/clouddrivedb`

4. Run the application

- Make sure you add a connection to `mongodb://localhost:27017` on Compass
- Make sure you're in the project root and run `npm run dev`
- Open <http://localhost:5173/> on your browser

This starts the backend on <http://localhost:3000> and the frontend on <http://localhost:5173> at the same time. See *requirements.txt* in the repository for the exact commands used to set up the project and install every dependency from scratch.

User Manual

1. **Register/Login:** First, create an account with a username (3-25 characters) and a password (minimum of 8 characters, with at least one lowercase letter, one uppercase letter, one number, and one symbol). Log in with the same credentials after registering.
2. **Messages:** Success and error messages appear throughout the app to help users. Mainly on error pages and after actions like share, revoke, login, and register.
3. **Drive:** After logging in, you land on your Drive.
 - You can create a new text document by clicking the *Create* button. You can upload an image by choosing a file and then clicking the *Upload Image* button (titles are defaulted to the original image file name).
 - You can search your documents, sort them, and page through them with the *Show More* button at the bottom
4. **Navigation Bar:** The top left features a button to return to the drive if you're in the recycle bin or profile page. The top right features buttons to toggle between light and dark mode, to go to your profile, to log out, and to switch the UI's language (EN & FI).
5. **Profile:** Upload a profile picture from the Profile page. If none is set or the uploaded file is not in an image format, a default letter avatar (based on the username) is given instead.
6. **Creating & Editing Documents:** Click a document's title to open it. Text documents feature a text editor while image documents are shown directly. Remember to click the *Save* button after editing the text contents or title!
7. **Sharing:** On the owner's view of a document, you can give edit access to another user by username, or toggle a public link that lets anyone view the document (read-only) without having to log in. Owners can revoke edit access by username as well.
8. **Downloading:** Both text documents and image documents can be downloaded as a PDF. Text documents downloaded as a PDF lose any formatting done in the text editor, while images downloaded as a PDF take up the whole PDF page. When image documents that can move (GIF, webp, etc) are downloaded as a PDF, only the first frame is downloaded. Images can also be downloaded directly in their original format.
9. **Recycle bin:** Deleting a document first moves it to the recycle bin (accessible via the trash icon on the Drive page), from which it can be restored or permanently deleted.

API Endpoints

Auth

Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	Log in, returns a JWT
GET	/api/auth/me	Get the logged-in user's info
PUT	/api/auth/profile-picture	Upload a profile picture

Documents

Method	Endpoint	Description
GET	/api/documents	Get all documents owned by or shared with the user
POST	/api/documents	Create a new document
GET	/api/documents/trash	Get all documents in the recycle bin
GET	/api/documents/:id	Get a single document (owner, editor, or public view)
PUT	/api/documents/:id	Edit a document's title and/or content
DELETE	/api/documents/:id	Move a document to the recycle bin
PUT	/api/documents/:id/share	Give edit access to a user by their username
PUT	/api/documents/:id/revoke	Revoke edit access from a user by their username
PUT	/api/documents/:id/public	Toggle the public read-only link
PUT	/api/documents/:id/restore	Restore a document from the recycle bin
DELETE	/api/documents/:id/permanent	Permanently delete a document
POST	/api/documents/:id/clone	Clone an existing document
POST	/api/documents/upload-image	Upload an image as a new document
PUT	/api/documents/:id/lock	Claim the editing lock on a document
PUT	/api/documents/:id/unlock	Release the editing lock on a document

Known Limitations

- **Responsive testing:** I tested the application on various screen sizes on Chrome DevTools' Device Mode to get a clearer idea of how it would look and work on desktop and mobile environments. The smallest width I supported for was 375px. This is because it was the smallest preset device width in Device Mode on Chrome DevTools that I could find. Beyond this, the application should display correctly across most desktop and mobile screen sizes.
- **File Upload Validation:** No file validation was implemented for the process of uploading profile pictures. Uploading a non-image file causes it to revert to the default letter avatar rather than displaying an error. For image documents in the drive, uploading an unsupported file type (for example a PDF) shows a placeholder graphic with a message below it which informs the user the uploaded image isn't a supported file type and also lists various supported image formats.
- **PDF Formatting:** Any formatting applied in the document editor (like bolding, underlining, italicizing, bullet points, etc.) will not translate over when a text document is downloaded as a PDF.
- **Animated Images:** Moving images (like GIFs and animated WebPs) downloaded from the drive will still have their original animation, since the download is just a direct copy of the original uploaded file. A downloaded file's animation depends on which app opens it. Windows built-in *Photos* app, for example, shows the animation in GIF files but doesn't for animated WebP files. However, opening either file in a browser animates both correctly. So if a downloaded image looks still, try to open it in a browser before assuming something is broken. Additionally, downloading an animated image as a PDF only includes its first frame, since PDFs cannot contain animated content.
- **Editing Lock:** The editing lock is tied to the browser tab being open. If a user's device loses power or crashes without closing the tab normally, the lock could remain until manually cleared by the user who was editing it last (since `beforeunload/sendBeacon` can't guarantee delivery in every possible scenario).

Miscellaneous Notes

- Visiting a page that requires authentication (Drive, Trash, Profile) while logged out shows an informational message with a guide on how to navigate back to the login page, rather than redirecting automatically.
- Uploaded images (profile pictures and image documents) are stored on the server's file system, with their file path stored in MongoDB.
- Date and time displayed in tooltips follow the currently selected UI language (English or Finnish), and not the visitor's device settings like *Created* and *Last Updated* do.
- The dark mode settings reset to light mode when the tab is closed (uses sessionStorage)

Acknowledgements

- Light & dark mode and recycle bin icons taken from [react-icons](#)
- Document text editor made possible by [Quill](#)
- Document → PDF converter made possible by [jsPDF](#)
- Responsive design + UI components from [Bootstrap](#)
- Internationalization powered by [i18next](#) and [react-i18next](#)

Declaration of AI Usage

AI Systems Used: ChatGPT & Gemini

- **Date Tooltips:** I was inspired by a feature I noticed on Reddit where, when a user hovers over the date of when a post or comment was created, a black box appears showing the exact day, date, and time. I wanted this feature to appear whenever I hovered over the dates for columns *Created*, *Last Updated*, and *Deleted At*. I asked AI what this feature was called and was told it's called a "tooltip" and suggested I include a *title* component inside the opening tag of each table data cell I want it in. I wrote the values for *title* myself in *Drive.tsx*, *Trash.tsx*, and *Navigation.tsx*.
- **Password Regex:** I needed to use a regex for validating password inputs on the login and register pages to display error messages. In the early stages of the project, I decided to follow the same password requirements from *Exercise 8*, which honestly made writing out the regex quite troublesome. I eventually gave up and asked AI to give me the regex that fit the criteria I had. The regex was used in *Register.tsx* and *Login.tsx*.
- **Quill & jsPDF:** I asked AI for suggestions on which WYSIWYG editor I should choose and how I would go about downloading documents as a PDF. It suggested I use *Quill* for the editor and *jsPDF* for downloading as PDF's. I read the documentation and implemented them by myself.
- **Dark Mode Problems:** When toggling between dark and light mode, I ran into the issue of everything except the text editor icons not switching to white or a light color when in dark mode. AI suggested I add the lines from 115-125 in my *index.css* file and it worked.
- **Sort Documents By Name:** When implementing the function for sorting documents in the drive based on a selection, I initially had a roundabout way of sorting the documents alphabetically (it worked, but I overcomplicated it). I asked AI if there was a simpler way to sort by name. It suggested I use the *localeCompare* method in my if statement in line 128 of *Drive.tsx*.
- I had a bit of trouble translating some error messages from English to Finnish whenever the swap language button was toggled. Some of my messages needed to include a dynamic value like a username, but the course content only covered static string examples like `{t("Revoke")}`, where nothing inside the parentheses changes. I asked AI, and it helped me understand `i18next`'s interpolation syntax, where a placeholder like `{{username}}` is written inside the string and the real value is passed in separately. I used this method in lines 421 and 475 in *DocumentEditor.tsx*.
- **JWT Payload Decoding on Frontend:** I needed to know which user was currently logged in on the frontend to check permissions in *DocumentEditor.tsx*, like whether someone was the owner or an editor of a doc. The course only covered creating and verifying JWTs on the backend, not reading one on the frontend. I asked AI how to get my own user id out of the token, and it showed me a JWT's payload is just base64 encoded, so it can be decoded with `JSON.parse(atob(token.split(".")[1]))`. I used this in the beginning of *DocumentEditor.tsx*.
- **Editing lock design:** I had a bit of trouble preventing two users from editing the same document at the same time. I asked AI for a general approach, and it suggested storing the id of whoever currently has a document open in the database, then checking it whenever someone opens a document, and then releasing it when they leave. I implemented the actual claim and release routes in *documents.ts* myself, along with the *beforeunload* and *sendBeacon* logic in *DocumentEditor.tsx* to release the lock if the tab is closed.